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Title

**Hamiltonian-Encoded Quantum LSTM for
Learning Higher-Order Dependencies in
Multivariate Time-Series Forecasting**

REVIEW REPORT – I

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comparison, not a statistically validated result.

Abstract

Multivariate time-series forecasting in domains such as power-grid load management, transformer thermal monitoring, and climate modelling increasingly relies on recurrent architectures capable of capturing long-range and higher-order temporal dependencies. Classical LSTM networks, and recent hybrid Quantum LSTM (QLSTM) models, embed input data into variational quantum circuits (VQCs) through shallow per-timestep angle-embedding layers, which constrains their representational capacity and limits their ability to model continuous, coupled temporal dynamics. This project proposes a Hamiltonian-Encoded Quantum LSTM (H-QLSTM), in which the input at each time step parameterizes a learnable Hamiltonian operator $H(x, \theta)$, and the recurrent quantum state evolves under a Trotterized time-evolution operator $U = \exp(-iH\Delta t)$ derived from that Hamiltonian, rather than through static rotation gates. This dynamical-systems-based encoding is intended to more naturally capture coupled, higher-order temporal dependencies and long-range correlations in multivariate sequences than existing angle-embedding QLSTM designs.

As a 30% checkpoint (Phase I) toward this goal, a classical LSTM baseline and an existing angle-embedding QLSTM baseline (a 4-gate variational quantum cell adapted from Kea et al., 2024) were implemented and trained on the Electricity Transformer Temperature (ETTh1) dataset. The QLSTM baseline, configured with 6 qubits and 2 variational layers over 24-hour lookback windows, achieved a test RMSE of 0.59 °C, MAE of 0.39 °C, and MAPE of 4.89% after an Optuna-based hyperparameter search and early-stopped training. Phase II will design, implement, and integrate the proposed Hamiltonian encoding into the QLSTM cell, extend evaluation to the Electricity Load Diagrams and Jena Climate datasets, and benchmark the H-QLSTM against both baselines. The expected contribution is a systematic framework for designing, optimizing, and benchmarking Hamiltonian-driven quantum recurrent models for multivariate time-series forecasting.

Keywords: Quantum Machine Learning; Quantum LSTM; Hamiltonian Encoding; Variational Quantum Circuits; Multivariate Time-Series Forecasting

1. Introduction

Sequential, physically-grounded systems — power grids, electrical transformers, and regional climate — generate multivariate time series whose future values depend on nonlinear, multi-scale, and higher-order couplings between past observations. Forecasting such series accurately supports load balancing, preventive transformer maintenance, and climate-risk planning. Recurrent neural networks, and quantum-enhanced variants of them, are the dominant modelling family for this task; this project examines how the input encoding used inside a quantum recurrent cell affects its ability to represent those dependencies.

1.1 Background

Classical Long Short-Term Memory (LSTM) networks use gated memory cells — input, forget, cell, and output gates — to mitigate the vanishing-gradient problem and retain information over long sequences. Their representational capacity is nonetheless bounded by classical linear-algebraic gate operations. Quantum machine learning models sequence data using variational quantum circuits (VQCs): parameterized rotation gates interleaved with entangling layers, trained by gradient descent via the parameter-shift rule. Existing Quantum LSTM (QLSTM) architectures replace each of the four classical LSTM gates with a small VQC, embedding the input into per-timestep gate rotations through angle embedding (data values mapped directly to rotation angles). Hamiltonian embedding, demonstrated for static data by Wang et al. (2025), instead encodes data into the coefficients of a Hamiltonian operator $H(x)$ and evolves the quantum state under the unitary $U = \exp(-iH\Delta t)$, giving continuous, physically-motivated dynamics in place of discrete gate rotations.

Defining “higher-order temporal dependency”. The phrase is used precisely here rather than as a slogan. Statistically, a target y has an order- p dependency on lagged inputs $x_{t-1}, \dots, x_{t-p}$ if y is independent of every strict subset of those lags and only their full joint interaction carries information about y — the k -wise-independence property of, e.g., the parity function; a linear function of many lags is still order-1 in this sense, so “higher order” is not the same as “long lookback window”. Dynamically, if inputs are encoded into a time-evolution generator $U(x,t) = \mathcal{T}\exp(-i\int H(x(t))dt)$, its Dyson/Magnus expansion produces terms such as $\int\int T[H(t_1)H(t_2)]dt_1dt_2$, whose n -th order term nonlinearly couples n distinct time points through the non-commutativity of H at different times — a fixed-ansatz VQC gate recurrence (the Phase I baseline) is Markovian by construction and has no algebraic access to these terms, which is exactly the gap the proposed Hamiltonian encoding targets.

1.2 Motivation

A gap exists between the static, shallow angle-embedding used in current QLSTM designs and the continuous, coupled temporal evolution that many physical time series actually follow. Embedding the input into the generator of the circuit's own time evolution — rather than into a fixed set of rotation gates — couples the quantum state's evolution directly to the input dynamics at every time step. This motivates a Hamiltonian-encoded QLSTM cell that is expected to model higher-order dependencies and longer-range temporal correlations more effectively than existing angle-embedding QLSTM cells, particularly in domains such as transformer thermal behaviour and climate systems where the underlying process is itself a continuous dynamical system.

1.3 Objectives

- 1 To implement and validate a classical LSTM and an existing angle-embedding QLSTM as baseline models for multivariate time-series forecasting.
- 2 To design a Hamiltonian-Encoded QLSTM (H-QLSTM) cell in which input features parameterize a learnable Hamiltonian and the recurrent state evolves under its time-evolution operator.
- 3 To implement the proposed H-QLSTM using PennyLane and PyTorch and integrate it into the existing preprocessing / training / evaluation pipeline.
- 4 To evaluate all three models on the ETTh1, Electricity Load Diagrams, and Jena Climate datasets using RMSE, MAE, and MAPE.
- 5 To compare the proposed H-QLSTM against the classical LSTM and existing QLSTM baselines, particularly with respect to longer lookback windows and higher-order dependency capture.

1.4 Scope

Included: design and implementation of a classical LSTM, an existing angle-embedding QLSTM, and the proposed H-QLSTM; simulated quantum circuit execution (PennyLane *default.qubit*) sized to remain classically tractable (approximately 6–8 qubits, 2–3 variational layers); evaluation on three public multivariate time-series datasets (ETTh1, Electricity Load Diagrams, Jena Climate). **Excluded:** execution on physical quantum hardware; real-time or streaming deployment; forecasting horizons beyond the windowed single/multi-step setting already validated in the Phase I checkpoint.

2. Related Work

Table 2.1 summarizes the principal work this project builds on and positions itself against: hybrid quantum-classical recurrent forecasting, Hamiltonian-based quantum data embedding, and quantum recurrent forecasting architectures more broadly.

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Kea et al. (2024) [1]	4-gate QLSTM cell: each classical LSTM gate replaced by a VQC with angle-embedded features and entangling layers.	Competitive with classical LSTM/GRU baselines on stock-price forecasting with fewer trainable parameters.	Data embedded via static per-timestep angle rotations; no explicit modelling of continuous temporal dynamics; single-domain (stock) evaluation.
2	Wang et al. (2025) [2]	Quantum Hamiltonian embedding: data reuploaded as coefficients of a Hamiltonian, state evolved via its time-evolution operator.	Richer expressivity and improved trainability (fewer barren-plateau issues) versus gate-angle embedding on image classification.	Applied to static image data; no recurrent/sequential architecture or temporal-dependency modelling.
3	Moon et al. (2025) [3]	QSegRNN: a segment-wise quantum recurrent unit for time-series forecasting.	Reduces circuit depth and parameter count versus standard quantum RNNs while improving forecasting accuracy on benchmark series.	Segment gating still relies on variational angle embedding; limited capacity for long-range, higher-order dependency modelling.
4	Hochreiter & Schmidhuber (1997) [4]	Classical LSTM: gated recurrent memory cell with input/forget/output gates.	Mitigates vanishing gradients; remains the standard classical baseline for sequence modelling.	Purely classical representational capacity; motivates hybrid quantum approaches for richer feature spaces.

Table 2.1: Summary of related work

Summary of Related Works table is mandatory and it should list the mainly referred documents for your work. Don't change the table column headings.

2.1 Research Gap

Existing hybrid QLSTM and quantum-RNN designs (Kea et al.; Moon et al.) embed data through shallow, static variational angle encodings within the gate circuits, which does not explicitly model continuous-time dynamics. Hamiltonian embedding (Wang et al.) has been shown to improve expressivity for static data but has not been integrated into a recurrent, sequential architecture. No systematic framework currently exists to design, optimize, and benchmark a Hamiltonian-driven QLSTM against classical and variational-quantum baselines for multivariate time-series forecasting with higher-order and long-range temporal dependencies — this is the gap this project addresses.

3. Problem statement

Classical and current hybrid quantum recurrent models struggle to efficiently capture higher-order temporal dependencies in complex nonlinear time series due to limited representational capacity and shallow data encodings. Quantum LSTM (QLSTM) frameworks have shown promise in exploiting quantum feature spaces for sequence learning, but existing designs primarily embed data via variational circuits without explicit Hamiltonian-driven dynamics to model continuous temporal evolution. A dynamical Hamiltonian encoding integrated within a QLSTM can embed data into the evolution operator itself, enabling richer modelling of temporal couplings and long-range dependencies. However, no systematic framework has been developed to design, optimize, and benchmark such Hamiltonian-based QLSTM architectures against classical and variational quantum baselines. Addressing this gap will advance quantum sequence learning and open pathways for scalable, dynamical-system-based temporal models.

Input: windowed multivariate time-series sequences (24-hour lookback windows) drawn from the ETTh1, Electricity Load Diagrams, and Jena Climate datasets, with cyclical time features and scaled numerical features.

Output: a forecast of the target variable (e.g. oil temperature for ETTh1) at the next timestep(s), together with RMSE, MAE, and MAPE against held-out ground truth.

Constraints: the quantum circuit must remain classically simulatable for benchmarking (bounded qubit count and circuit depth, PennyLane *default.qubit* on CPU); training must use standard gradient-based optimization (autodiff / parameter-shift); all three models (classical LSTM, existing QLSTM, proposed H-QLSTM) must be trained and evaluated under identical data splits, preprocessing, and metrics.

Criteria: success is measured on the real datasets by RMSE, MAE, and MAPE relative to the classical LSTM and existing QLSTM baselines, and — since low error on ETTh1 alone does not demonstrate order- p sensitivity — by sign-accuracy on a controlled order- p parity diagnostic (Section 5.2.4) that is provably unsolvable below the true interaction order p , giving a direct, swept measure of higher-order dependency capture rather than an inferred one.

4. Justification for the problem formulation

This chapter justifies the technical need for a Hamiltonian-encoded QLSTM, the feasibility of building and evaluating one within the project timeline, and the contribution it is expected to make.

4.1 Need for the Proposed Work

Quantum sequence models remain nascent, and no systematic Hamiltonian-based QLSTM benchmark currently exists in the literature. As quantum hardware and simulators mature, establishing whether a dynamically-motivated encoding meaningfully improves temporal-dependency modelling — versus simply adding circuit depth — is an open and practically important question for the quantum machine learning community.

4.2 Feasibility

Technical: the PennyLane + PyTorch stack required is already operational; a classical LSTM and an existing angle-embedding QLSTM baseline have been implemented, trained, and evaluated on ETTh1 as the Phase I checkpoint (Section 5.3), confirming the pipeline end-to-end. **Operational:** a CPU-tractable configuration (6 qubits, 2 layers, hidden size 8) has already been identified via Optuna hyperparameter search and trains in under an hour on a 3,500-row subsample. **Economic:** all datasets are public and free; no proprietary licences or paid compute are required for circuit simulation at this scale. **Schedule:** the remaining work — designing and implementing the Hamiltonian encoding, integrating it into the QLSTM cell, and benchmarking across three datasets — is scoped into Phase II (Chapter 6).

4.3 Expected Contribution

The anticipated contribution is a systematic framework for designing, optimizing, and benchmarking Hamiltonian-driven quantum recurrent models: (i) a Hamiltonian-Encoded QLSTM cell that embeds input into a learnable Hamiltonian's time-evolution operator; (ii) an open, reproducible benchmark of this cell against a classical LSTM and an existing angle-embedding QLSTM across three public multivariate time-series datasets; and (iii) a quantified assessment of whether Hamiltonian encoding improves higher-order and long-range temporal dependency modelling over existing quantum recurrent designs.

5. Methodology

The complete workflow spans data preprocessing, three parallel recurrent-model implementations sharing a common training/evaluation harness, and comparative benchmarking, as summarized below and shown in Figure 5.1.

5.1 Architectural Design for Proposed System

Raw multivariate series are cleaned, time-encoded, scaled, and windowed by the input layer. The processing layer hosts three interchangeable recurrent components trained and evaluated under an identical harness: the classical LSTM baseline, the existing angle-embedding QLSTM baseline, and the proposed H-QLSTM, whose cell encodes the input into a learnable Hamiltonian $H(x, \theta)$ and evolves the quantum state via the Trotterized operator $U = \exp(-iH\Delta t)$ in place of static rotation gates. The output layer applies a classical regression head to the recurrent state and reports RMSE, MAE, and MAPE.

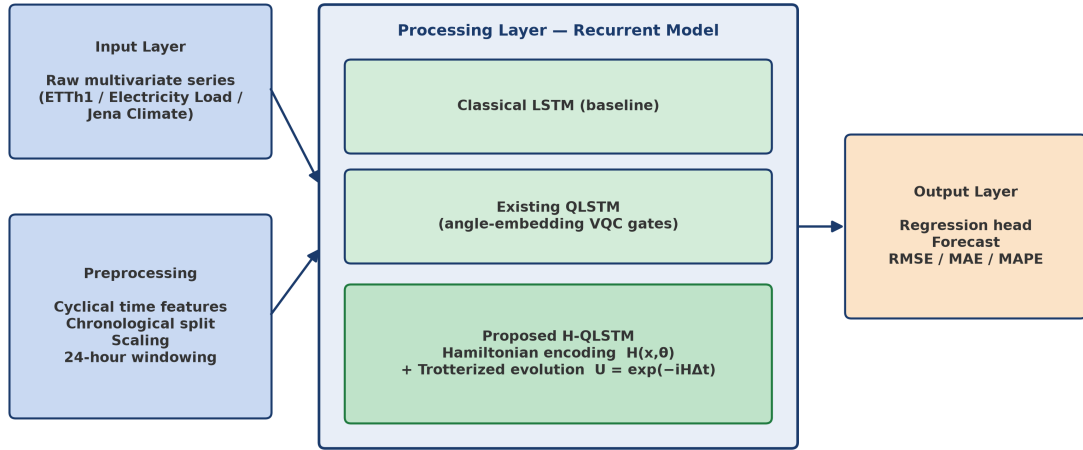


Figure 5.1: High-level architecture of the proposed system

5.2 Modules

5.2.1 Module 1: Data/Input Management

Loads ETTh1 (and, in Phase II, the Electricity Load Diagrams and Jena Climate datasets), derives cyclical time-of-day / day-of-week features, performs a chronological train/validation/test split, applies feature scaling, and constructs 24-hour lookback windows for sequence-to-one supervised learning (src/preprocessing.py).

5.2.2 Module 2: Core Processing

Implements the recurrent cells: the classical LSTM; the existing 4-gate angle-embedding QLSTM cell (src/qlstm_model.py); and, in Phase II, the proposed H-QLSTM cell in which each gate's VQC is replaced by a Hamiltonian-parameterized time-evolution block.

5.2.3 Module 3: User Interface and Output

run_pipeline.py orchestrates preprocessing, hyperparameter search, final training, and evaluation end-to-end, writing metrics, loss curves, and prediction-versus-actual plots to results/ for reporting.

5.2.4 Module 4: Testing and Evaluation

train.py implements the train/validation loop with early stopping; hyperparam_search.py runs an Optuna search over qubit count, layer depth, hidden size, and learning rate; utils.py computes RMSE, MAE, and MAPE used to benchmark all three models under identical conditions. synthetic_order_experiment.py provides a controlled capability probe that is independent of any real dataset: it constructs order- p parity tasks ($y = x_1 \times \dots \times x_p$ over i.i.d. Rademacher inputs), which are provably unsolvable from any strict subset of the p lagged inputs, and sweeps p to report sign-accuracy against chance (50%) for the QLSTM and a parameter-comparable classical LSTM — a direct, swept test of order- p capture that ETTh1 accuracy alone cannot provide, to be run against the H-QLSTM cell in Phase II once it is implemented.

5.3 Preliminary Results (Phase I Checkpoint)

The existing angle-embedding QLSTM baseline was trained on a 3,500-hour subsample of ETTh1 with 24-hour lookback windows, 6 qubits, and 2 variational layers, following an 8-trial Optuna hyperparameter search. A classical LSTM baseline (standard nn.LSTM gates, hidden size 8, matched to the QLSTM's selected configuration) was trained under the identical harness — same data split, scaling, windowing, optimizer, loss, and early-stopping protocol — for a direct comparison; it was not separately hyperparameter-searched, since at this scale it is cheap and is not itself the object of the comparison. Table 5.1 reports test-set performance for both; Figures 5.2 and 5.3 show the QLSTM's training/validation loss curve and predicted-versus-actual oil temperature, and Figure 5.4 overlays both models' forecasts against actual values.

Metric	QLSTM	Classical LSTM
Test RMSE (original units, °C)	0.5919	0.6316
Test MAE (original units, °C)	0.3909	0.4021
Test MAPE	4.89%	5.15%
Best validation MSE (scaled)	0.03926	0.05039
Epochs trained (early stop)	14 (patience 8)	28 (patience 8)
Training wall-clock time	52.3 min	1.6 s
Trainable parameters	n/a (VQC circuit)	681
Model configuration	6 qubits, 2 variational layers, hidden size 8, batch size 32, lr 0.01, sequence length 24 h	hidden size 8, batch size 32, lr 0.01, sequence length 24 h

Table 5.1: Phase I checkpoint — QLSTM vs. classical LSTM baseline test performance on ETTh1. QLSTM test RMSE is ~6.7% lower; this is a single-seed, single-configuration comparison, not a statistically validated result.

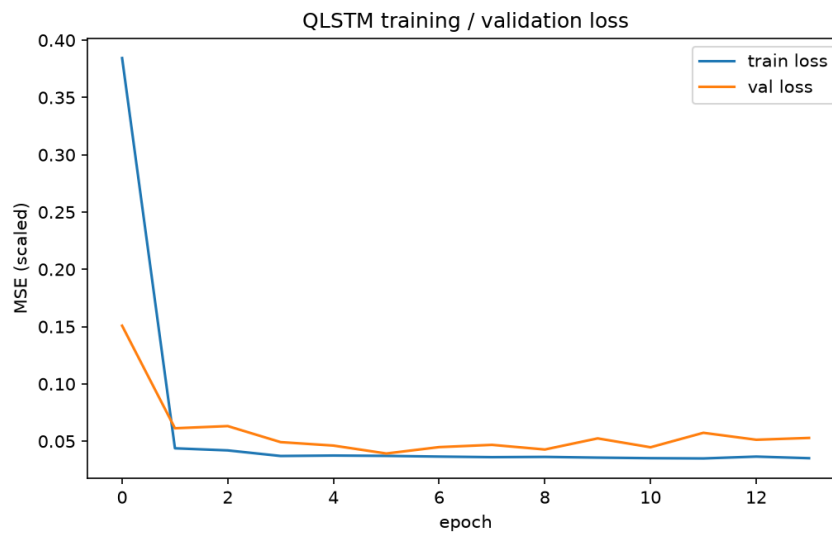


Figure 5.2: Training/validation loss curve — existing QLSTM baseline (Phase I checkpoint)

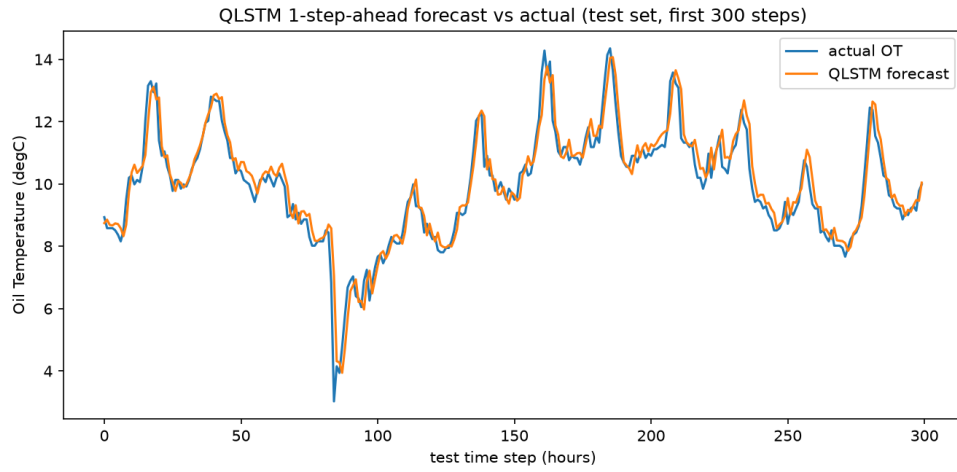


Figure 5.3: Predicted vs. actual oil temperature — existing QLSTM baseline (Phase I checkpoint)

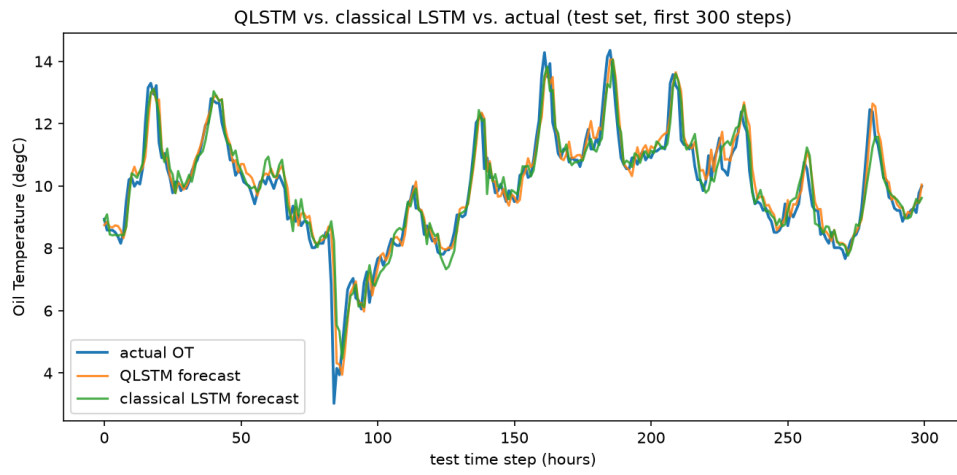


Figure 5.4: QLSTM vs. classical LSTM vs. actual oil temperature on held-out test data (Phase I checkpoint)

6. Project timeline chart

Figure 6.1 shows the two-phase project schedule. Phase I (August–December 2026) covers baseline development and the design of the proposed architecture; Phase II (January–June 2027) covers implementation, integration, optimization, and final benchmarking of the H-QLSTM.

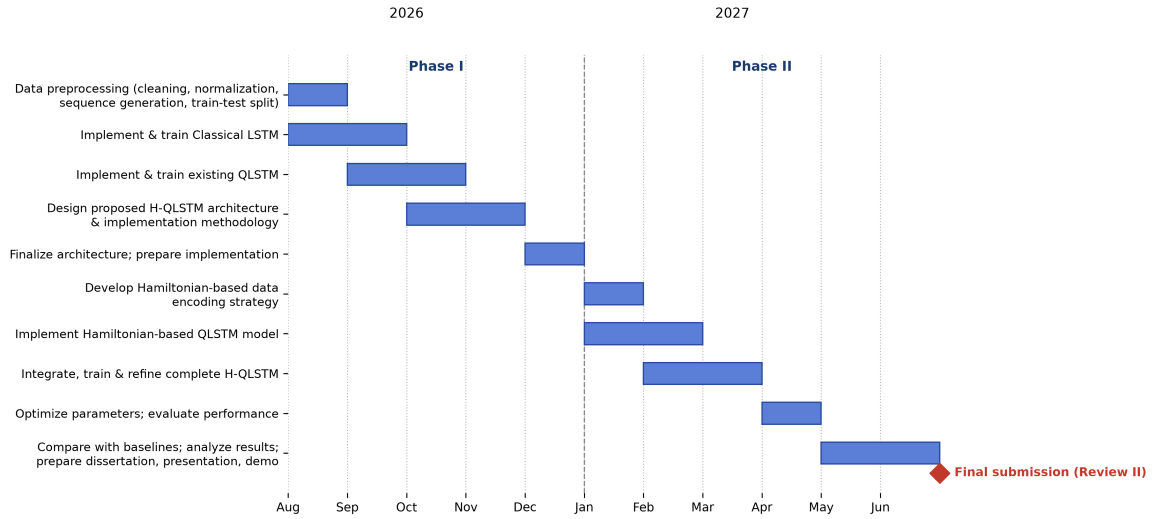


Figure 6.1: Project timeline

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